

Soluciones de Bases de Datos

Richard Guanoluisa

Bases de Datos II

Pontificia Universidad Católica
del Ecuador



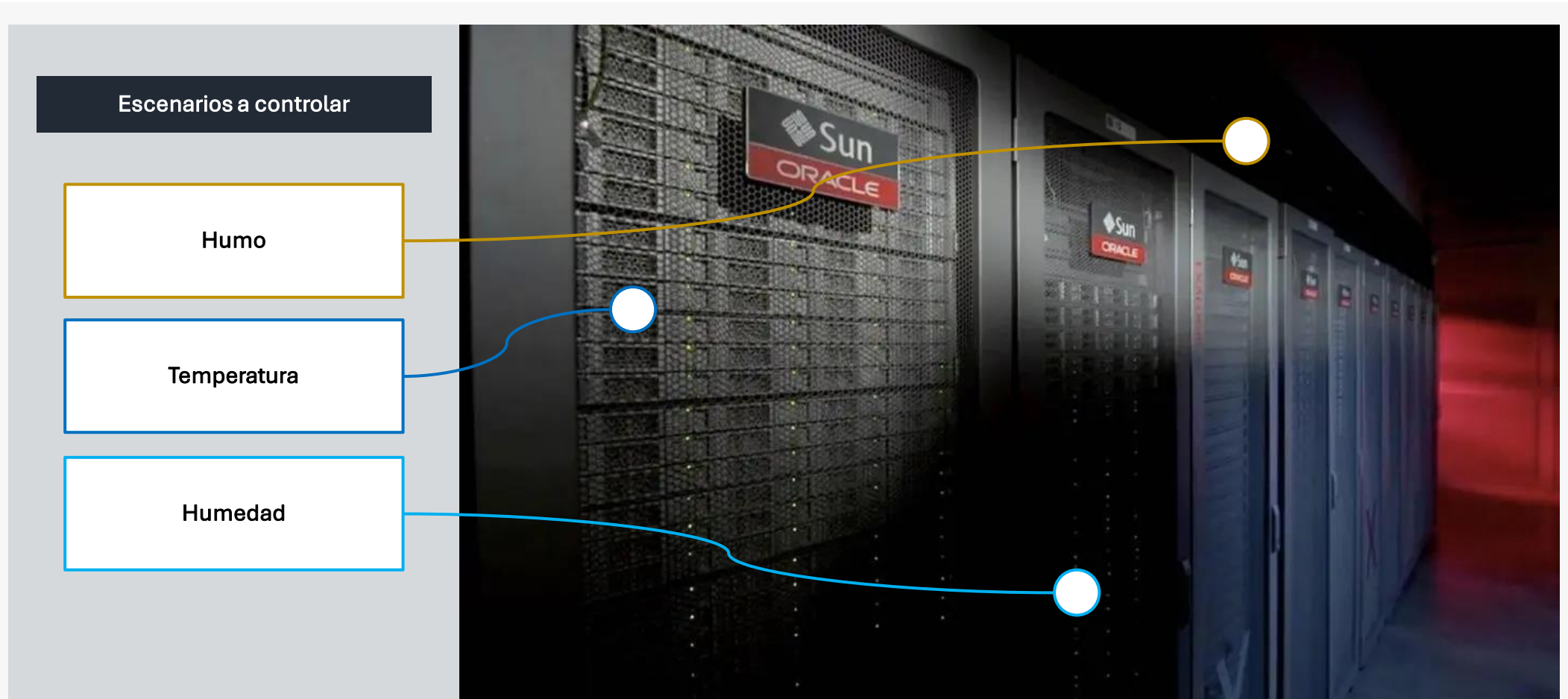
NoSQL - IoT

Las bases de datos NoSQL están diseñadas para manejar grandes volúmenes de datos heterogéneos y de alta frecuencia, como los generados por sistemas IoT. Su modelo flexible permite almacenar variables continuas y discretas sin esquemas rígidos, facilitando la escalabilidad, el análisis en tiempo real y la evolución del sistema sin interrupciones.



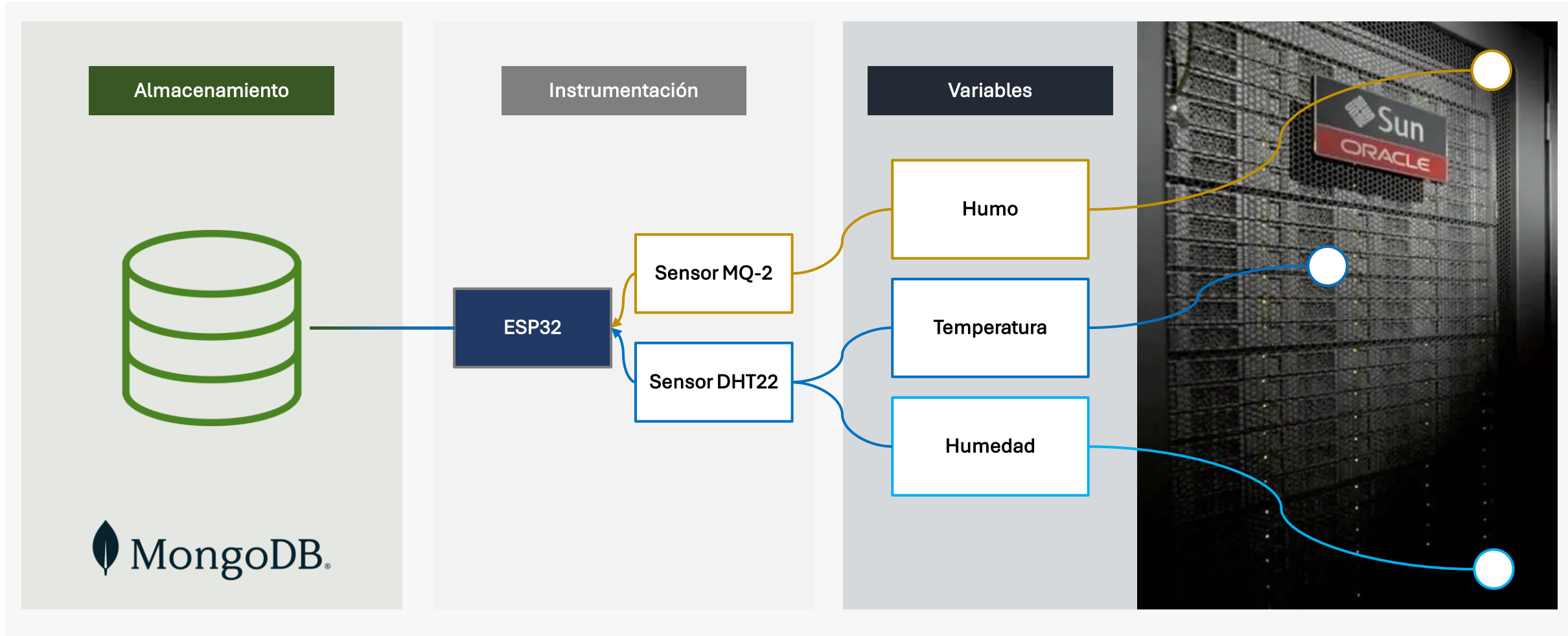
NoSQL - IoT

En ambientes críticos, es indispensable tener un ambiente controlado, las variables físicas deben medirse y recolectarse



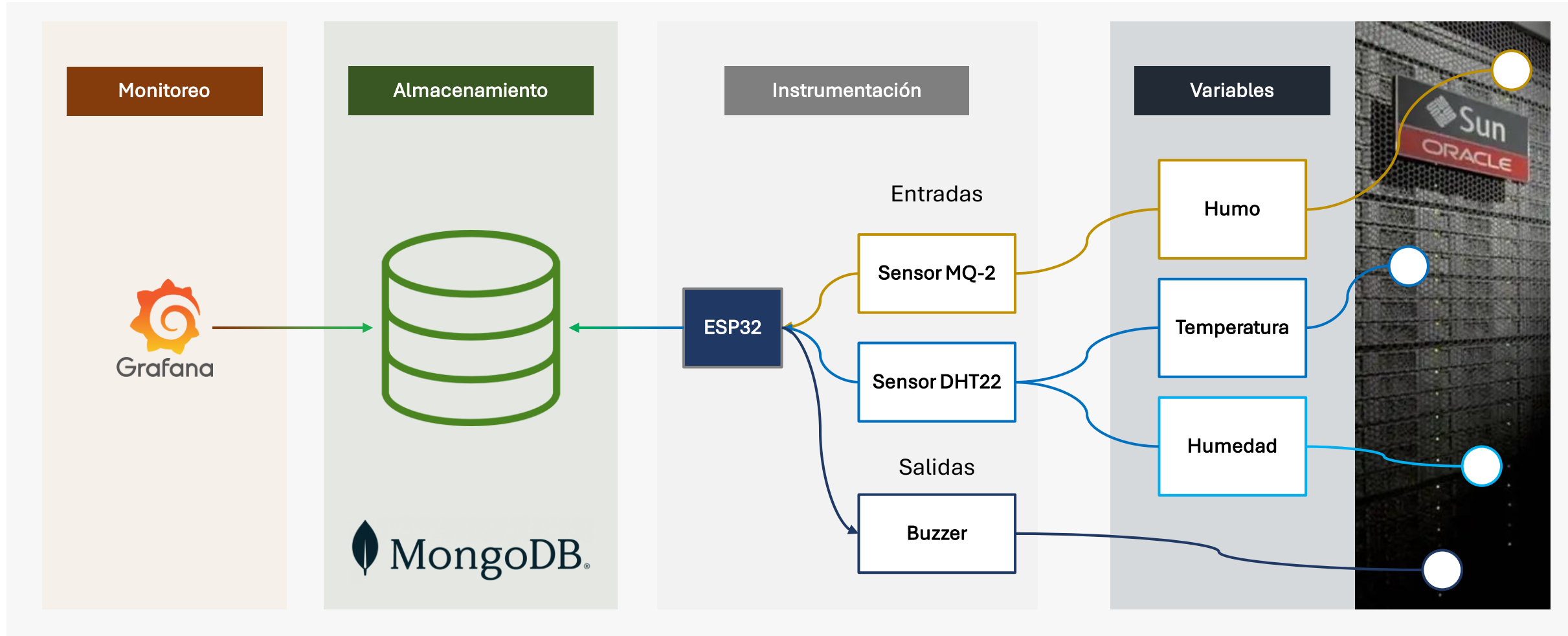
NoSQL - IoT

El tipo de dato define el moto de base de datos a implementar



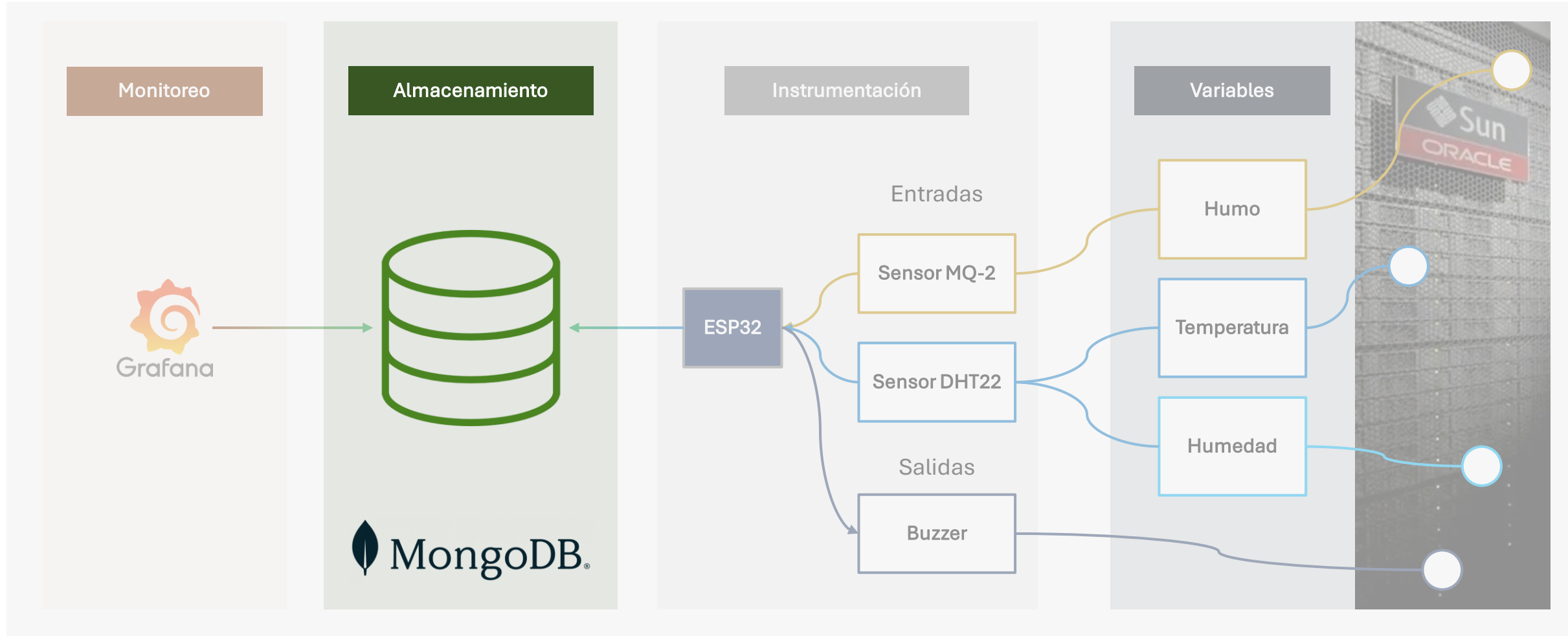
NoSQL - IoT

Un ecosistema completo permite monitorear y alertar las condiciones de mi sistema

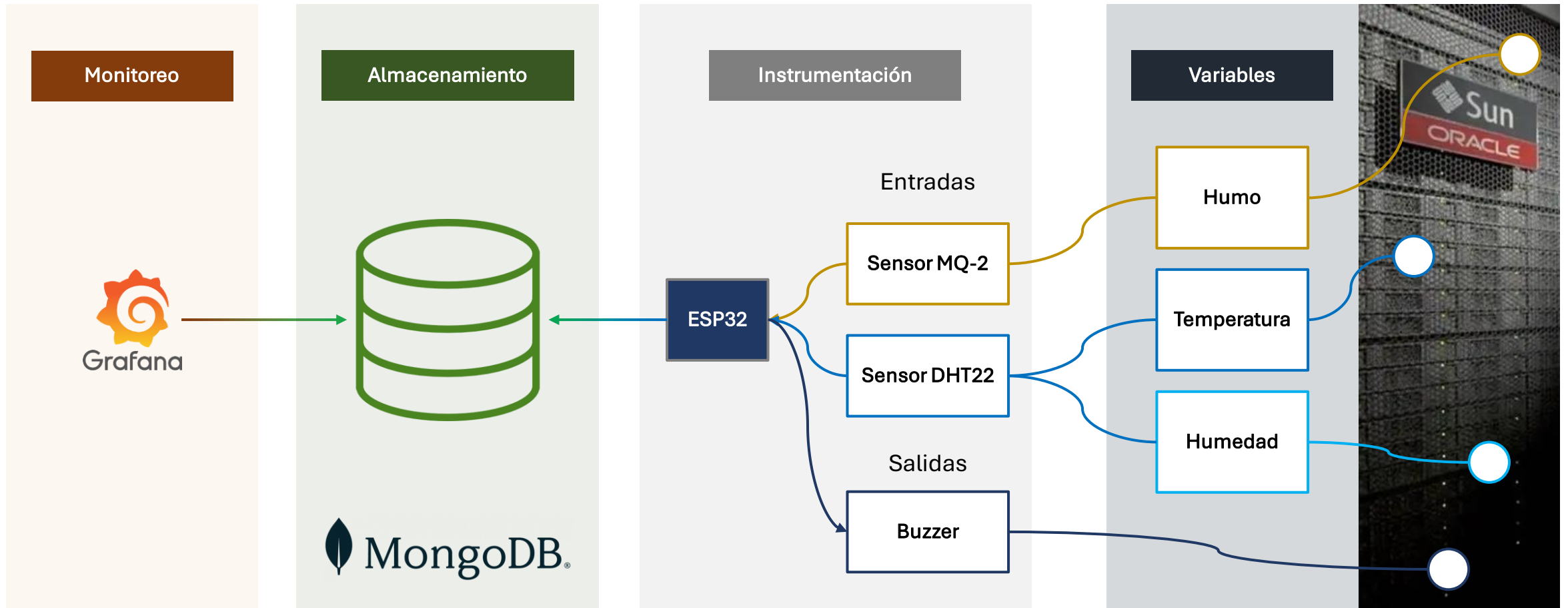


NoSQL - IoT

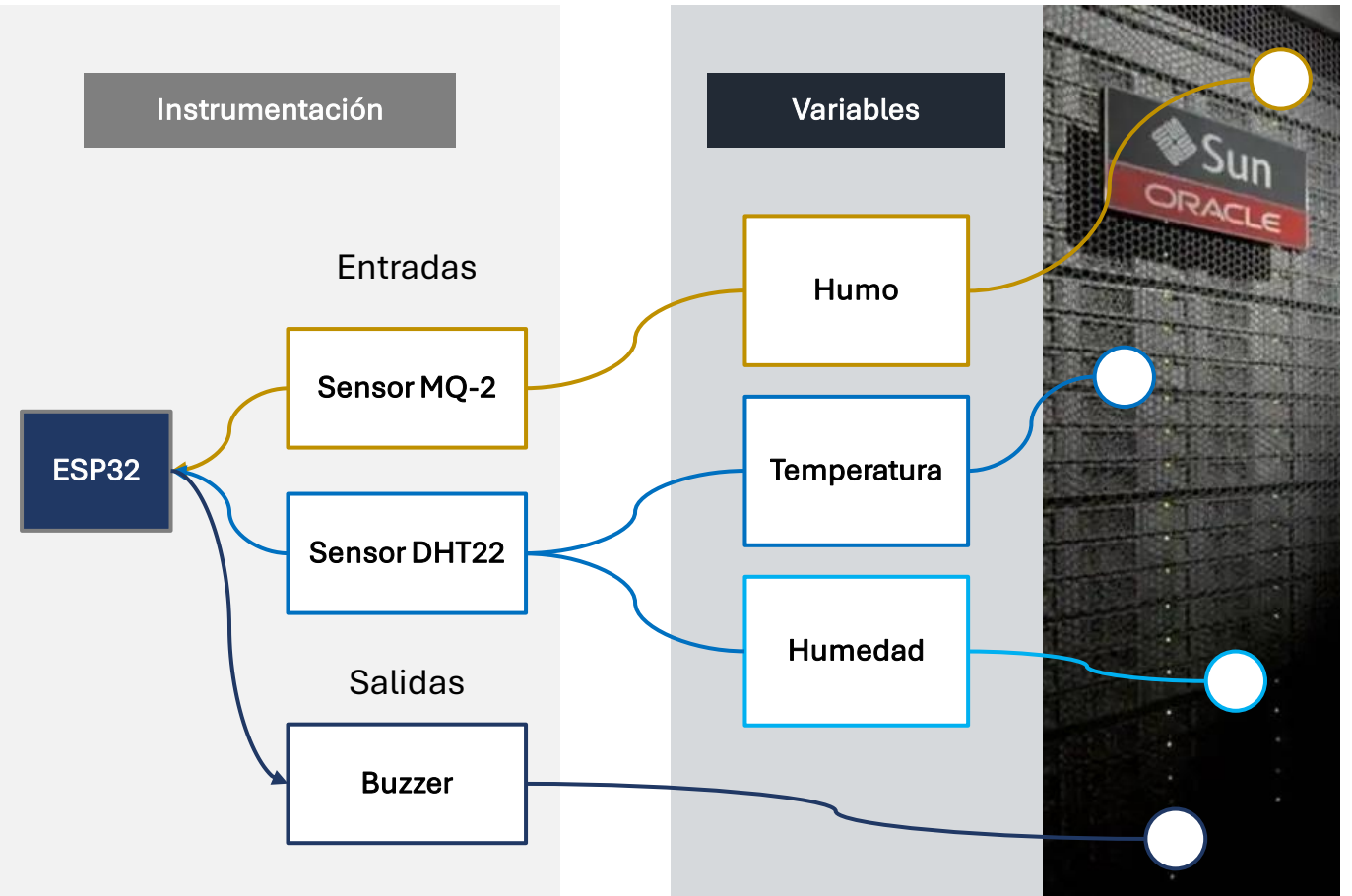
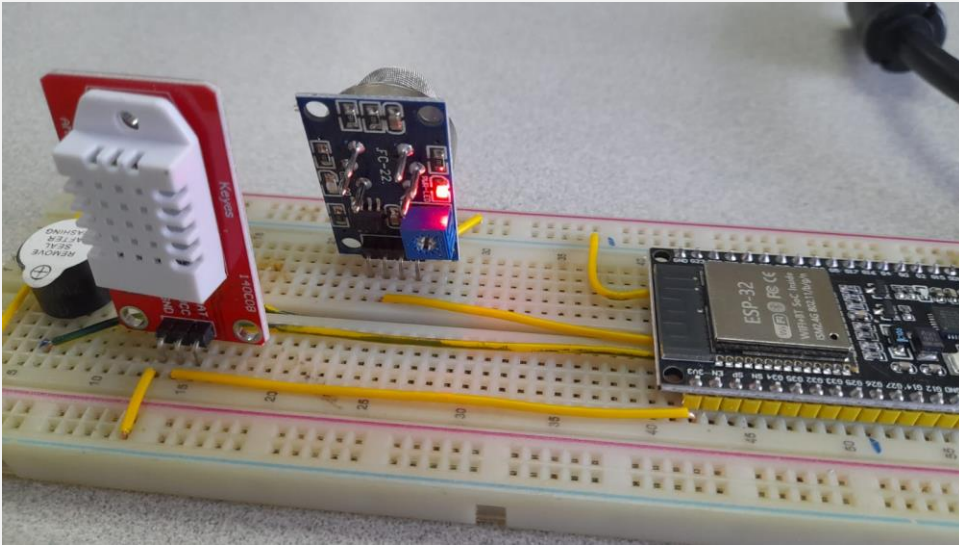
En este escenario, el motor de base de datos corresponde a una de las partes críticas ya que centraliza la información



Demostración - NoSQL - IoT



Demostración - NoSQL - IoT



Demostración - NoSQL - IoT

```
Insertado (10s): 21.4 65.7 0
Insertado (10s): 21.4 65.6 0
Insertado (10s): 21.5 65.5 0
Insertado (10s): 21.5 65.5 0
Insertado (10s): 21.5 65.5 0
Insertado (10s): 21.5 65.5 0
Insertado (10s): 21.5 65.6 0
Insertado (10s): 21.5 65.5 0
Insertado (10s): 21.5 65.5 0
Insertado (10s): 21.5 65.5 0
Insertado (10s): 21.5 65.5 0
Insertado (10s): 21.5 65.6 0
Insertado (10s): 21.5 65.4 0
Insertado (10s): 21.5 65.4 0
Insertado (10s): 21.5 65.3 0
Insertado (10s): 21.5 65.2 0
Insertado (10s): 21.6 65.3 0
Insertado (10s): 21.6 65.3 0
Insertado (10s): 21.6 65.3 0
Insertado (10s): 21.6 65.3 0
Insertado (10s): 21.6 65.2 0
Insertado (10s): 21.6 65.3 0
Insertado (10s): 21.6 65.2 0
Insertado (10s): 21.6 65.2 0
Insertado (10s): 21.6 65.1 0
Insertado (10s): 21.6 65.1 0
```

Connections Edit View Collection Help

Compass

{ } My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

iot

readings

local

readings

localhost:27017 > iot > readings

Documents 334 Aggregations Schema Indexes 2 Validation

Generate query

Explain

Reset

Find

Options

Project { field: 0 }

Sort { timestamp: -1 }

Collation { locale: 'simple' }

Index Hint { }

Max Time MS 60000

Skip 0 Limit 0

ADD DATA

UPDATE

DELETE

EXPORT DATA

EXPORT CODE

25 1 - 25 of 688

_id: ObjectId('697e1fc58fb26ee4cla73f2d')

deviceId: "esp32-01"

timestamp: 2026-01-31T15:29:09.703+00:00

temperature: 21.6

humidity: 65.1

smoke: 0

_id: ObjectId('697e1fbb8fb26ee4cla73f2c')

deviceId: "esp32-01"

timestamp: 2026-01-31T15:28:59.674+00:00

temperature: 21.6

humidity: 65.1

smoke: 0

_id: ObjectId('697e1fb18fb26ee4cla73f2b')

deviceId: "esp32-01"

timestamp: 2026-01-31T15:28:49.644+00:00

temperature: 21.6

humidity: 65.2

smoke: 0

_id: ObjectId('697e1fa78fb26ee4cla73f2a')

deviceId: "esp32-01"

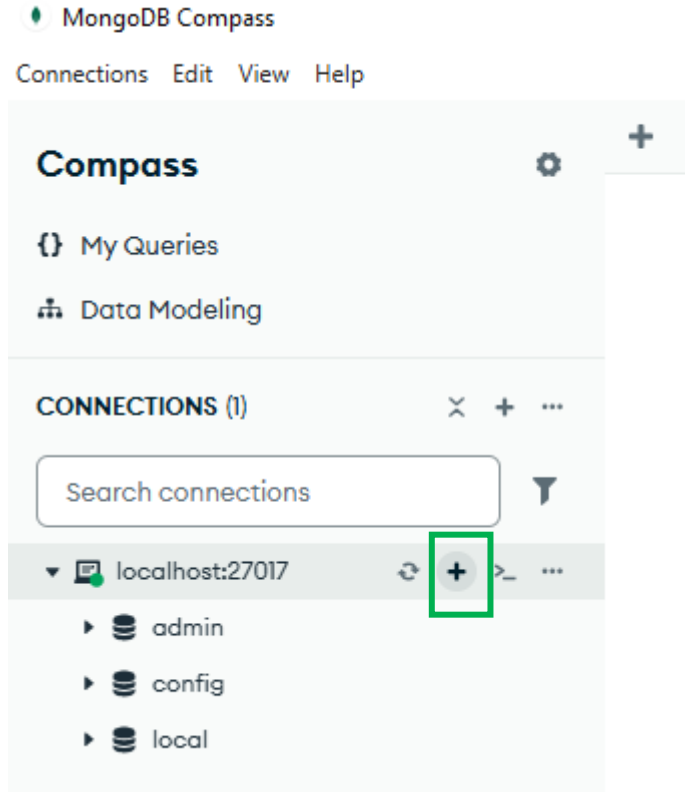
timestamp: 2026-01-31T15:28:39.614+00:00

temperature: 21.6

humidity: 65.2

smoke: 0

Procedimiento - NoSQL - IoT



Create Database

Database Name

iot

Collection Name

readings

☐ Time-Series

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

> Additional preferences (e.g. Custom collation, Clustered collections)

Cancel

Create Database

Procedimiento - NoSQL - IoT

MongoDB Compass - localhost:27017/iot.readings

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

iot

readings

local

readings

localhost:27017 > iot > readings

Open MongoDB shell

Documents 0

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT DATA

EXPORT CODE

25

0 - 0 of 0

Refresh

Previous

Next

Dropdown

Menu

JSON

Grid



This collection has no data

It only takes a few seconds to import data from a JSON or CSV file.

Import data

Procedimiento - NoSQL - IoT

Insertar dato de prueba

MongoDB Compass - localhost:27017/iot.readings

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

iot

readings

local

readings

localhost:27017 > iot > readings

Documents 0

Aggregations

Schemas

Type a query: { field: 'value'

ADD DATA

UPDATE

DELETE

Import JSON or CSV file

Insert document

Insert Document

To collection iot.readings

VIEW

```
1 {
2   "deviceId": "esp32-01",
3   "timestamp": {
4     "$date": "2026-01-31T01:00:00Z"
5   },
6   "temperature": 18.8,
7   "humidity": 72.4,
8   "smoke": 0
9 }
10
```

Cancel

Insert

Procedimiento - NoSQL - IoT

Insertar dato de prueba

MongoDB Compass - localhost:27017/iot.readings

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

iot

readings

local

readings

localhost:27017 > iot > readings

Documents 0

Aggregations

Schemas

Type a query: { field: 'value'

ADD DATA

UPDATE

DELETE

Import JSON or CSV file

Insert document

Insert Document

To collection iot.readings

VIEW

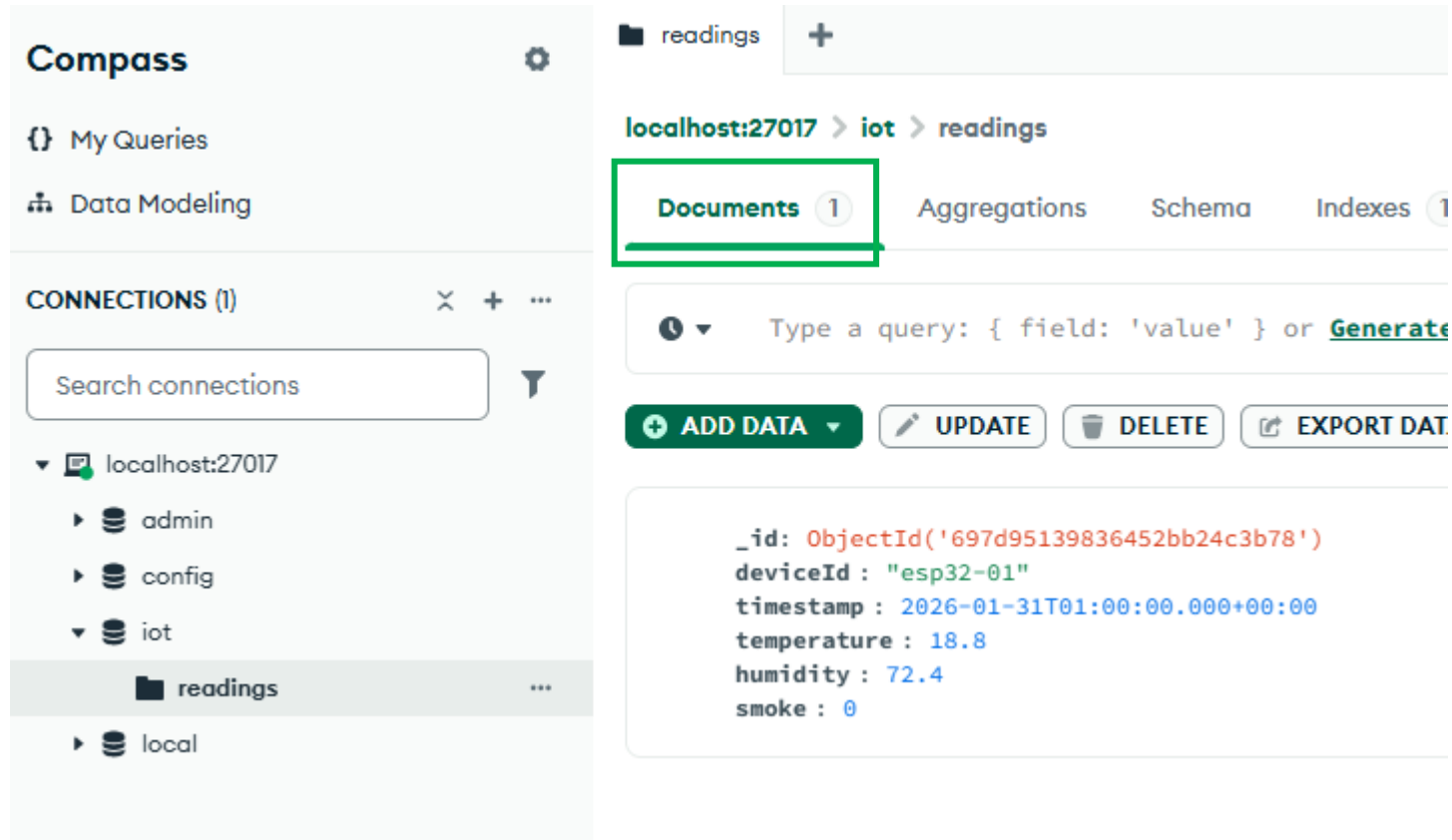
```
1 {
2   "deviceId": "esp32-01",
3   "timestamp": {
4     "$date": "2026-01-31T01:00:00Z"
5   },
6   "temperature": 18.8,
7   "humidity": 72.4,
8   "smoke": 0
9 }
10
```

Cancel

Insert

Procedimiento - NoSQL - IoT

Insertar dato de prueba



The screenshot shows the MongoDB Compass interface. On the left, the 'Connections' panel shows a connection to 'localhost:27017' with a tree view containing 'admin', 'config', 'iot', and 'readings' (selected). The main panel shows the 'readings' collection in the 'iot' database. The 'Documents' tab is selected and highlighted with a green box. Below the tabs, there is a query bar with a clock icon and the text 'Type a query: { field: 'value' } or [Generate](#)'. Below the query bar are four buttons: 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT DATA'. Below the buttons, a sample document is displayed:

```
{
  "_id": ObjectId('697d95139836452bb24c3b78'),
  "deviceId": "esp32-01",
  "timestamp": "2026-01-31T01:00:00.000+00:00",
  "temperature": 18.8,
  "humidity": 72.4,
  "smoke": 0
}
```

Procedimiento - NoSQL - IoT

Crear índice

MongoDB Compass - localhost:27017/iot.readings

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

iot

readings

local

readings

localhost:27017 > iot > readings

Documents 1

Aggregations

Schema

Indexes 1

Validation

Create Index

Refresh

VIEWING

Name & Definition

Type

Size

Usage

Properties

↓ _id_

REGULAR

20.5 kB

4 (since Sat Jan 31 2026)

UNIQUE

_id ↑

Procedimiento - NoSQL - IoT

Crear índice

Create Index

iot.readings

Index fields

timestamp 1 (asc) +

Options

☐ Create unique index

A unique index ensures that the indexed fields do not store duplicate values; i.e. enforces uniqueness for the indexed fields.

☒ Index name

Enter the name of the index to create, or leave blank to have MongoDB create a default name for the index.

idx_timestamp

Optional

☐ Create TTL

TTL indexes are special single-field indexes that MongoDB can use to automatically remove documents from a collection after a certain amount of time or at a specific clock time.

Cancel

Create Index

Procedimiento - NoSQL - IoT

Insertar dato de prueba (con índice)

Insert Document

To collection `iot.readings`

VIEW



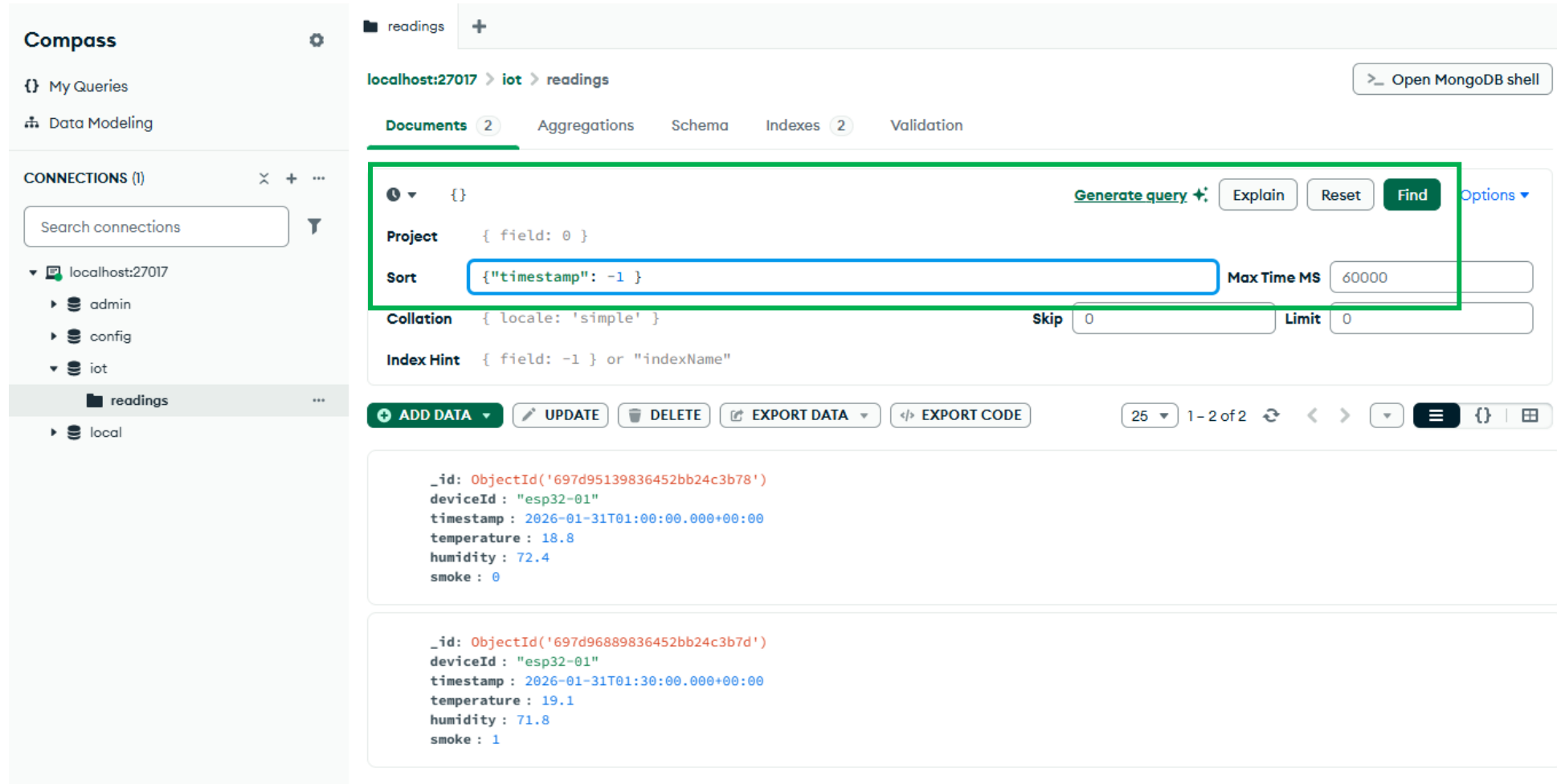
```
1 {  
2   "deviceId": "esp32-01",  
3   "timestamp": {  
4     "$date": "2026-01-31T01:30:00Z"  
5   },  
6   "temperature": 19.1,  
7   "humidity": 71.8,  
8   "smoke": 1  
9 }  
10 |
```

Cancel

Insert

Procedimiento - NoSQL - IoT

Insertar dato de prueba (con índice)



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel shows a connection to 'localhost:27017' with databases 'admin', 'config', and 'iot'. The 'readings' collection is selected. The main panel shows the 'Documents' tab for the 'readings' collection. A query is entered in the 'Sort' field: `{ "timestamp": -1 }`. The 'Max Time MS' is set to 60000. The 'Skip' and 'Limit' fields are both set to 0. The 'Find' button is highlighted. Below the query, two documents are displayed:

```
_id: ObjectId('697d95139836452bb24c3b78')
deviceId: "esp32-01"
timestamp: 2026-01-31T01:00:00.000+00:00
temperature: 18.8
humidity: 72.4
smoke: 0
```

```
_id: ObjectId('697d96889836452bb24c3b7d')
deviceId: "esp32-01"
timestamp: 2026-01-31T01:30:00.000+00:00
temperature: 19.1
humidity: 71.8
smoke: 1
```

Procedimiento - NoSQL - IoT

Crear proyecto de comunicación

```
C:\Users\rdgua>node -v
v22.17.1

C:\Users\rdgua>npm -v
10.9.2

C:\Users\rdgua>cd C:\Users\rdgua\OneDrive\Documentos\PUCE\20252 DB2\NoSQL-IoT\iot-api

C:\Users\rdgua\OneDrive\Documentos\PUCE\20252 DB2\NoSQL-IoT\iot-api>cd
C:\Users\rdgua\OneDrive\Documentos\PUCE\20252 DB2\NoSQL-IoT\iot-api

C:\Users\rdgua\OneDrive\Documentos\PUCE\20252 DB2\NoSQL-IoT\iot-api>npm init -y
Wrote to C:\Users\rdgua\OneDrive\Documentos\PUCE\20252 DB2\NoSQL-IoT\iot-api\package.json:

{
  "name": "iot-api",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

Procedimiento - NoSQL - IoT

Crear proyecto de comunicación

npm install express mongodb

```
added 77 packages, and audited 78 packages in 15s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
```

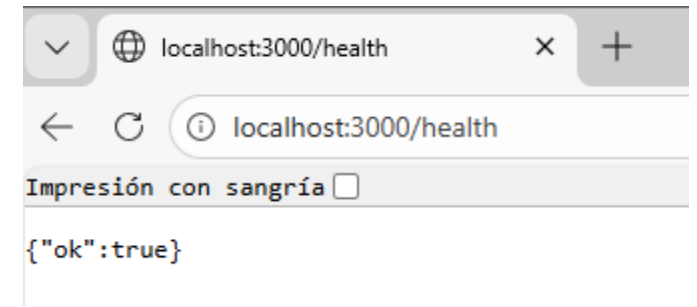


Este equipo > Documentos > PUCE > 20252 DB2 > NoSQL-IoT > iot-api >			
	Nombre	Estado	Fecha de mo
ido	node_modules		31/01/2026 0:
	package.json		31/01/2026 0:
:	package-lock.json		31/01/2026 0:
tos	server.js		31/01/2026 0:

Procedimiento - NoSQL - IoT

Crear proyecto de comunicación

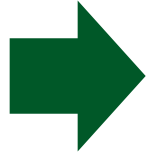
```
C:\Users\rdgua\OneDrive\Documentos\PUCE\20252 DB2\NoSQL-IoT\iot-api>node server.js  
API escuchando en http://localhost:3000
```



Procedimiento - NoSQL - IoT

Nuevo cmd

```
C:\Users\rdgua>curl -X POST http://localhost:3000/api/readings ^  
¿Más? -H "Content-Type: application/json" ^  
¿Más? -d '{"deviceId":"esp32-01","temperature":19.2,"humidity":70.5,"smoke":0}'  
{"insertedId":"697d99dc28493c53ae072482"}  
C:\Users\rdgua>
```



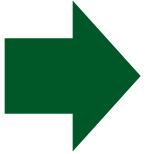
The screenshot displays the MongoDB Compass application. On the left, a sidebar shows the database structure: 'localhost:27017' contains collections 'admin', 'config', 'iot', and 'readings'. The 'readings' collection is selected and expanded, showing a list of documents. The main panel on the right displays three documents from the 'readings' collection. Each document contains fields for '_id', 'deviceId', 'timestamp', 'temperature', 'humidity', and 'smoke'. The third document is highlighted with a grey background.

Document	_id	deviceId	timestamp	temperature	humidity	smoke
1	ObjectId('697d95139836452bb24c3b78')	esp32-01	2026-01-31T01:00:00.000+00:00	18.8	72.4	0
2	ObjectId('697d96889836452bb24c3b7d')	esp32-01	2026-01-31T01:30:00.000+00:00	19.1	71.8	1
3 (Selected)	ObjectId('697d99dc28493c53ae072482')	esp32-01	2026-01-31T05:57:48.452+00:00	19.2	70.5	0

Procedimiento - NoSQL - IoT

Instalar librería Serial en Node

npm install serialport



```
C:\Users\rdgua>cd C:\Users\rdgua\OneDrive\Documentos\PUCE\20252 DB2\NoSQL-IoT\iot-api  
  
added 22 packages, and audited 100 packages in 8s  
  
36 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities
```

Instalar MongoDB



MongoDB

ANUNCIO

¡Presentamos MongoDB 8.0, el MongoDB más rápido de la historia! [Leer más >](#)

< >



Productos ▾

Recursos ▾

Soluciones ▾

Empresa ▾

Precios



🌐 Esp ▾

Support

Iniciar sesión

Prueba gratuita

| MONGODB ATLAS

A los desarrolladores
les encanta.
Diseñado para
aplicaciones
modernas.

atlas_vector_search.js

```
db.movieEmbeddings.aggregate([\n  {\n    \"$vectorSearch\": {\n      \"Index\": \"sample_movie_index\",  
      \"queryVector\": [0.09234, 0.49358, ...],  
      \"path\": \"plot_embedding\",  
      \"numCandidates\": 100,  
      \"limit\": 3,  
      \"filter\": {\n        \"year\": {\"$gt\": 1995}\n      }\n    }\n  }\n]);
```



¿Quiere ver si Mongo
adecuada para su pr



MBot • 3m

Explore pr



[MongoDB: la base de datos moderna líder mundial | MongoDB](#)

MongoDB



Productos ^

Recursos v

Soluciones v

Empresa v

Precios

PLATAFORMA



Atlas

Desarrolle en una plataforma de datos para desarrolladores

PLATAFORMA DE SERVICIOS



Base de Datos

Implemente una base de datos multicloud



Búsqueda

Brinde experiencias de búsqueda interactivas



Vector Search

Diseñe apps inteligentes con gen AI



Stream Processing

Integrar MongoDB y Kafka

AUTOGESTIONADO



Enterprise Advanced

Ejecute y gestione MongoDB usted mismo



Community Edition

Desarrolle a nivel local con MongoDB

HERRAMIENTAS



Compass

Trabaje con datos de MongoDB en una interfaz gráfica de usuario



Integraciones

Conéctese a servicios de terceros



Relational Migrator

Migre a MongoDB con confianza

Desarrolle con MongoDB Atlas

Comience gratis en minutos

[Regístrese](#)

Pruebe Enterprise Advanced

Desarrolle con MongoDB en las instalaciones

[Descargar](#)

Pruebe Community Edition

Explore la versión más reciente de MongoDB

[Descargar](#)

Ver todos los productos

Explora nuestra suite completa para desarrolladores

MongoDB 8.0

Nuestra versión más rápida hasta la fecha

MongoDB Community Edition

The source-available, free-to-use version of MongoDB's document database. Ideal for learning, experimenting, and building applications in a self-managed environment.

[Download Community](#)[Explore fully managed MongoDB →](#)

MongoDB



MongoDB.

Products ▾

Resources ▾

Solutions ▾

Company ▾

Pricing



Support

Sign In

MongoDB Atlas

MongoDB Enterprise Advanced

MongoDB Community Edition

MongoDB Community Server

MongoDB Controllers for
Kubernetes Operator

MongoDB Search in
Community

Tools

SQL Interface

Give it a try with a [free, highly-available 512 MB cluster](#). or get started from your terminal with the following two commands:

```
$ brew install mongodb-atlas  
$ atlas setup
```



Version

8.2.4 (current)



Platform

Windows x64



Package

msi



Download

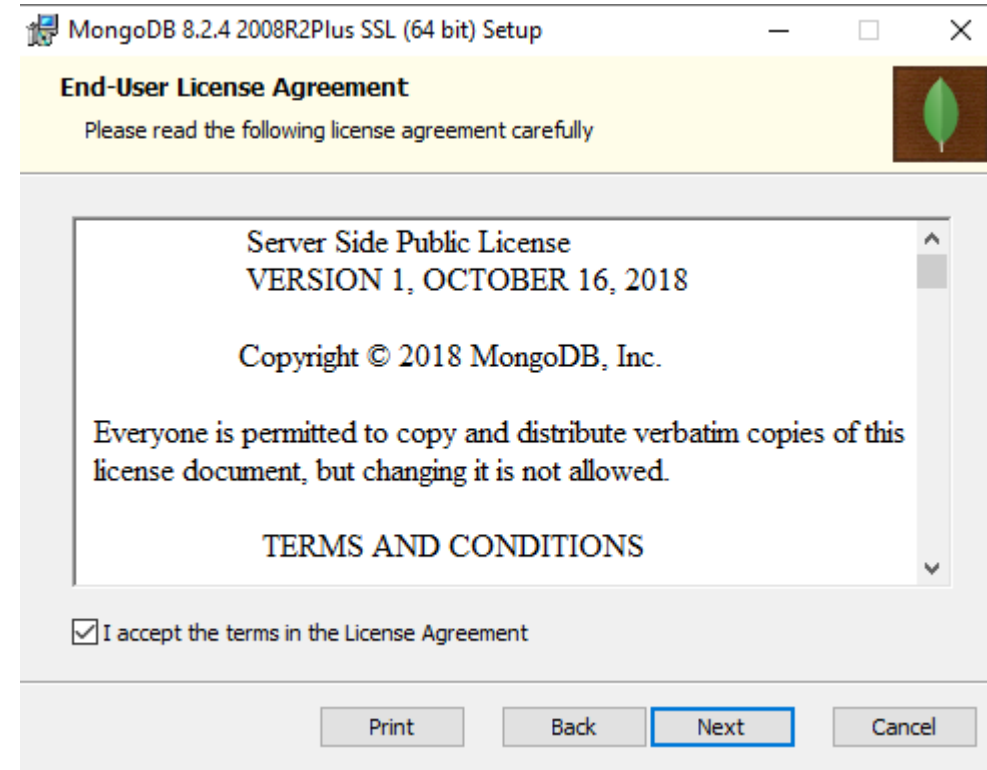
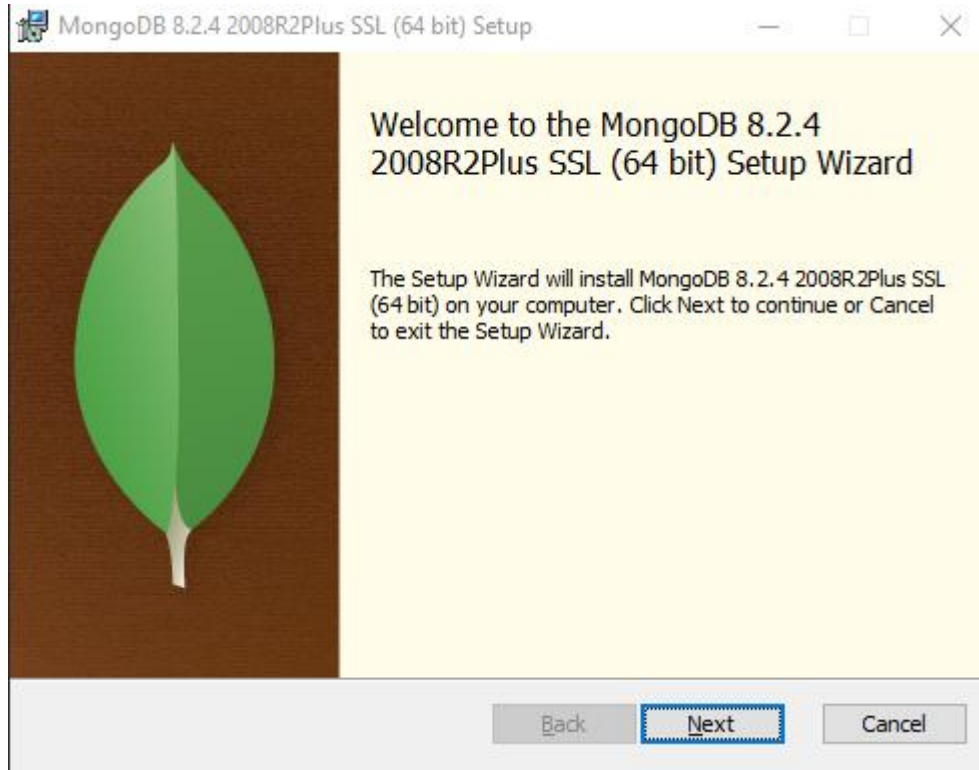


Copy link

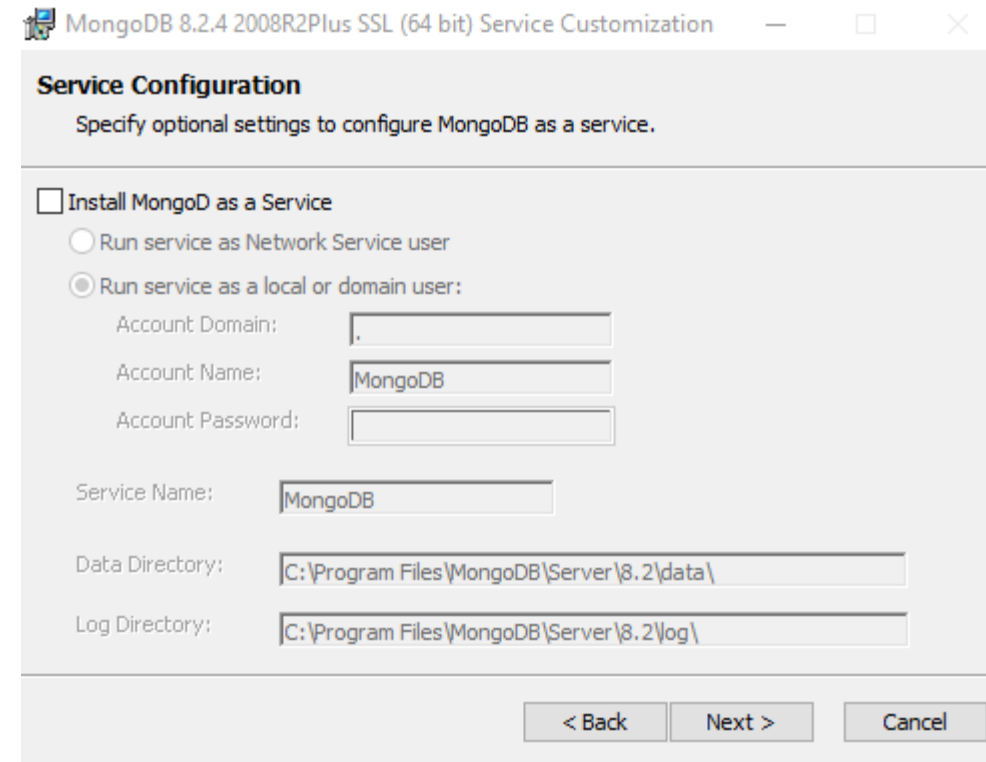
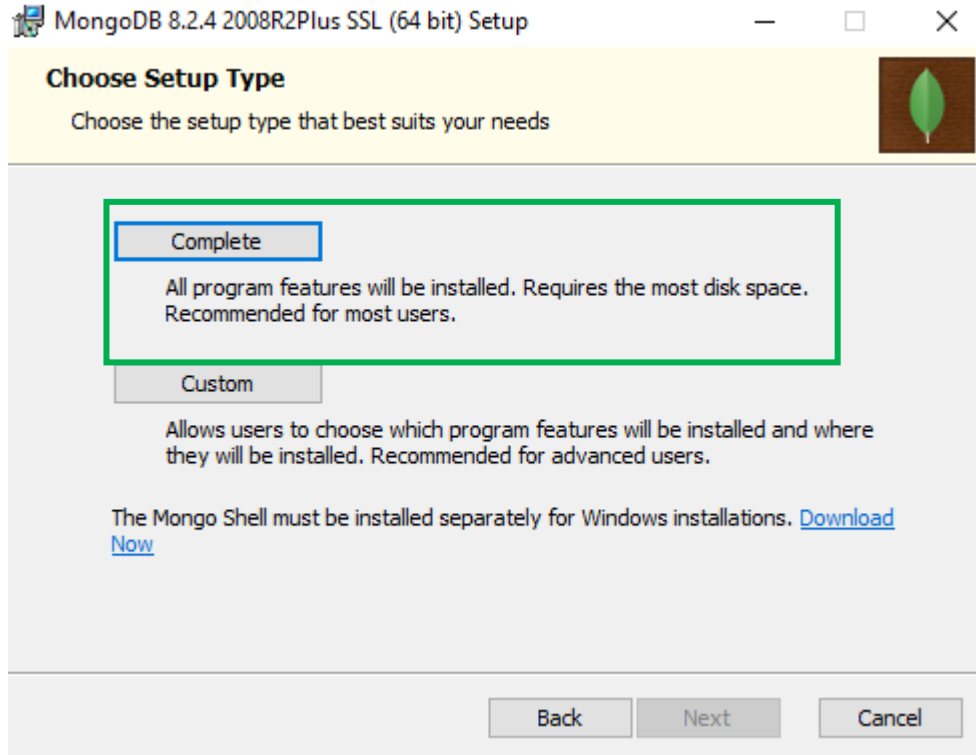
More Options



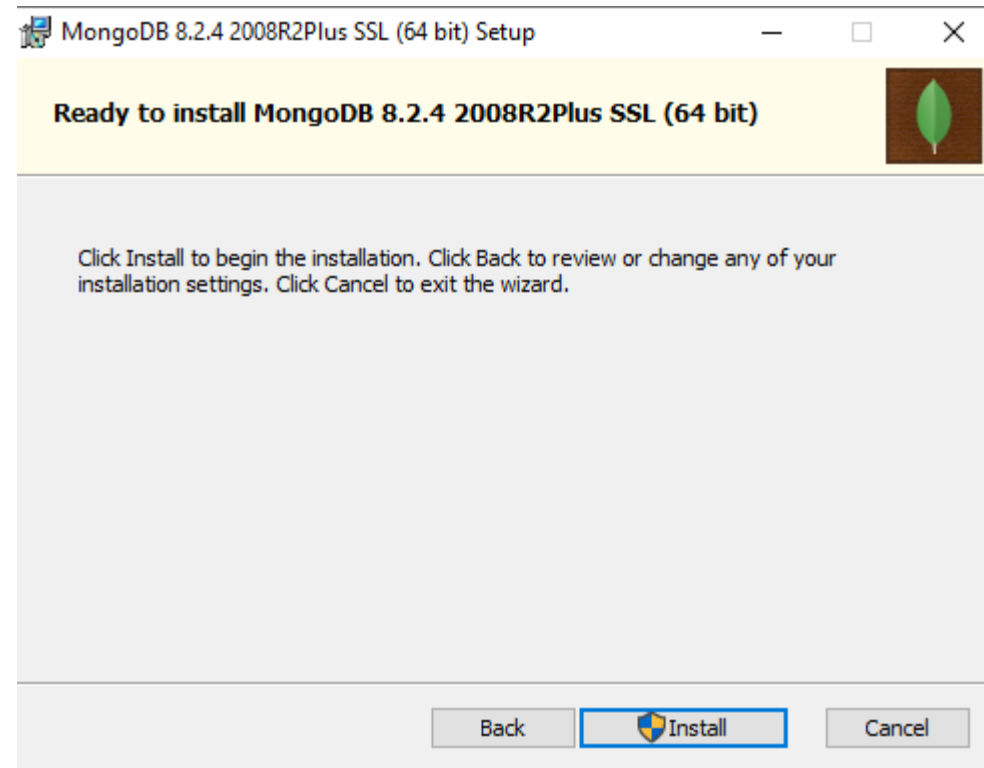
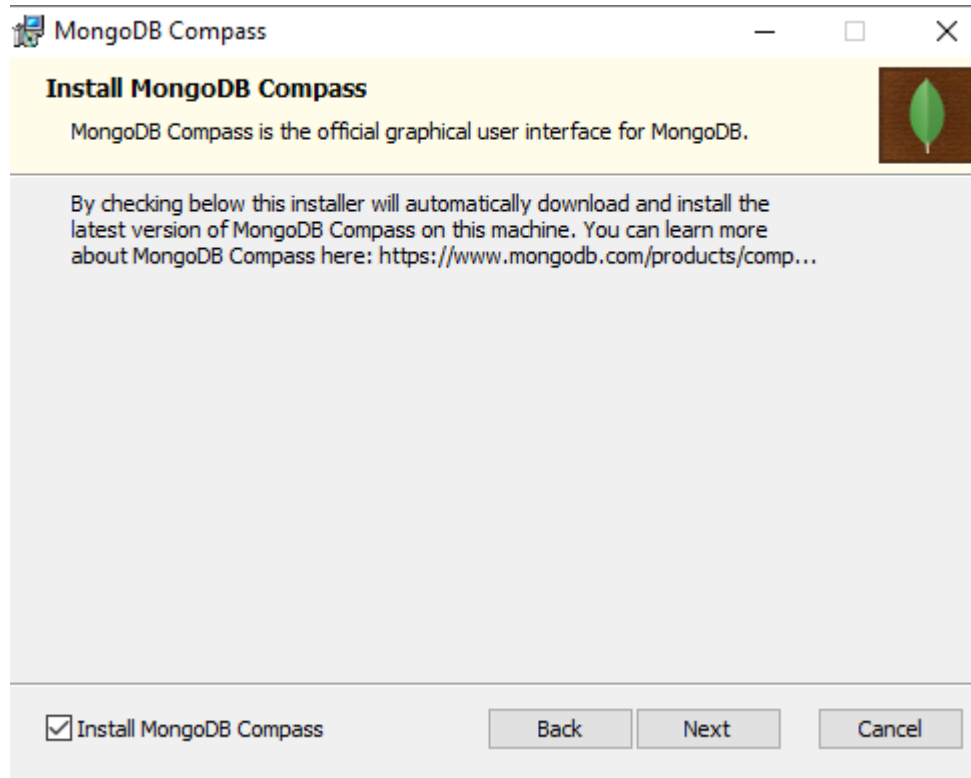
MongoDB



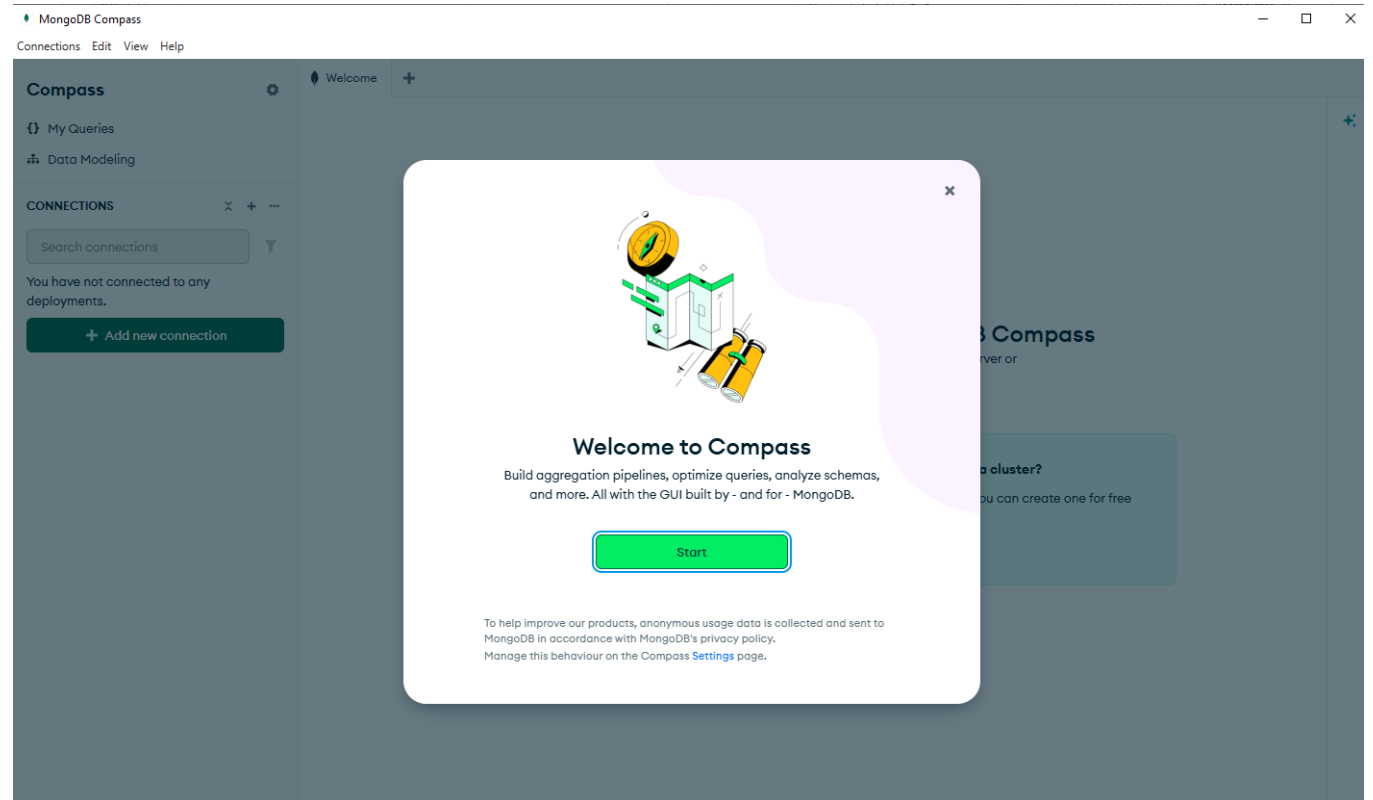
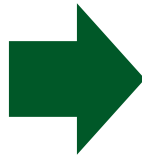
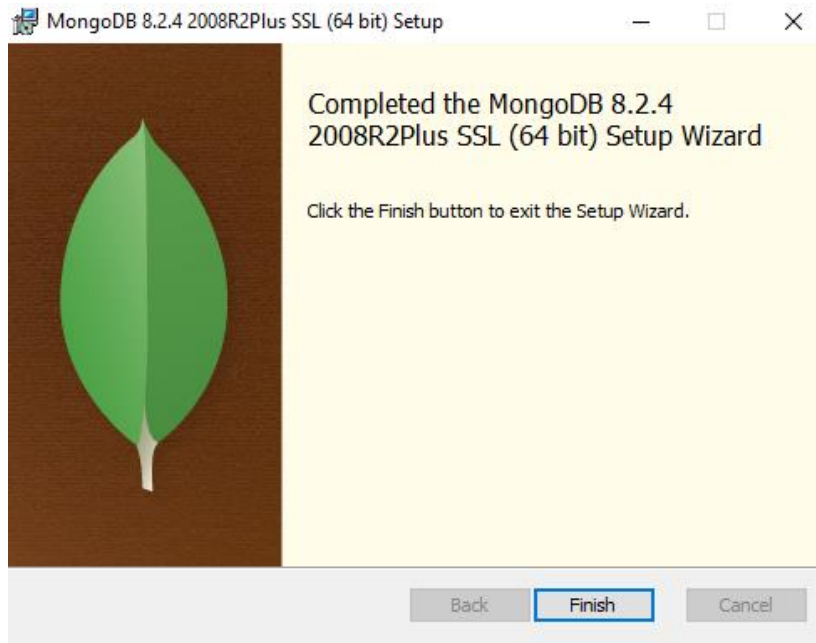
MongoDB



MongoDB

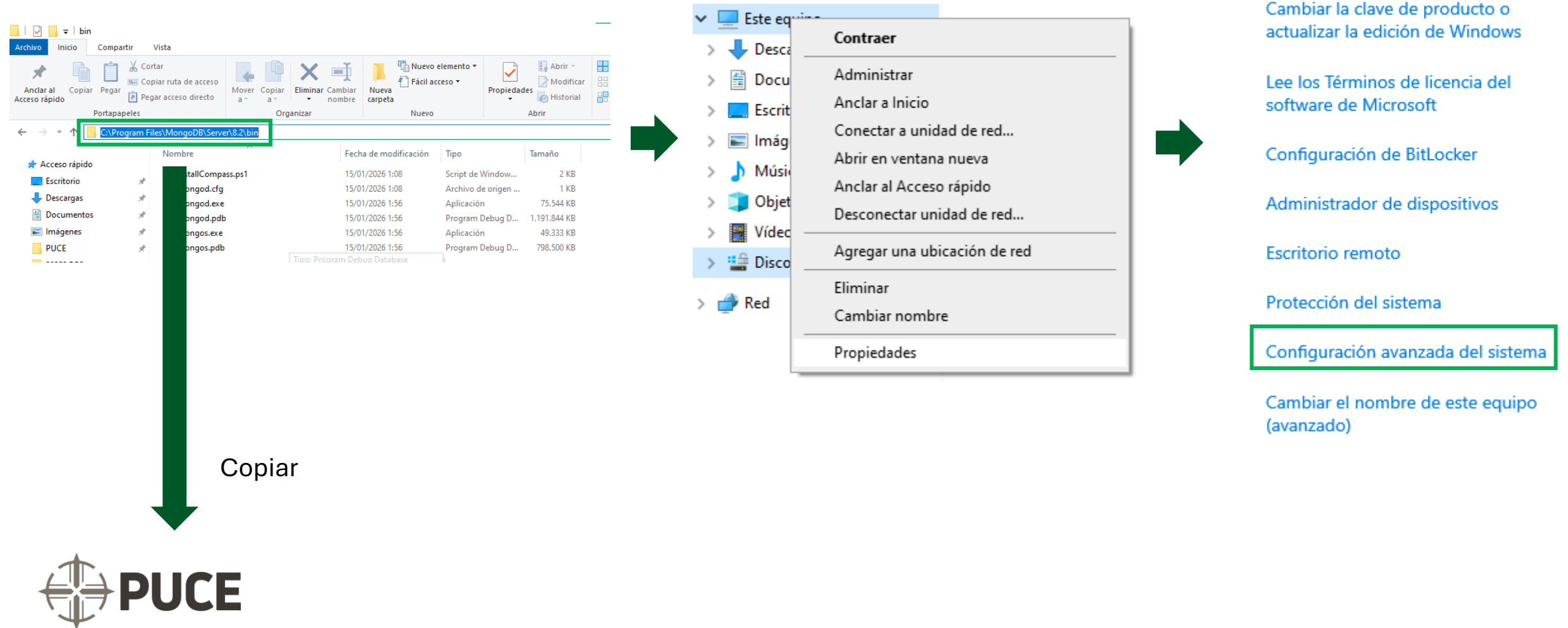


MongoDB



MongoDB

Añadir variable de entorno



MongoDB

Añadir variable de entorno

Tu equipo está supervisado

Propiedades del sistema

Nombre de equipo Hardware

Opciones avanzadas Protección del sistema Acceso remoto

Para realizar la mayoría de estos cambios, inicie sesión como administrador.

Rendimiento
Efectos visuales, programación del procesador, uso de memoria y memoria virtual

Perfiles de usuario
Configuración del escritorio correspondiente al inicio de sesión

Inicio y recuperación
Inicio del sistema, errores del sistema e información de depuración

Variables de entorno...

Aceptar Cancelar Aplicar

Procesador
RAM instalada
Tarjeta gráfica

Variables de entorno

Variables de usuario para rdgua

Variable	Valor
HADOOP_HOME	C:\hadoop
JAVA_HOME	C:\java
JD2_HOME	C:\Users\rdgua\AppData\Local\JDownloader 2
OneDrive	C:\Users\rdgua\OneDrive
OneDriveConsumer	C:\Users\rdgua\OneDrive
Path	C:\java\bin;C:\java\bin;C:\Program Files\Eclipse Adoptium\jdk-21.0...
PYSPARK_PYTHON	C:\Users\rdgua\AppData\Local\Programs\Python\Python39\pytho...

Nueva... Editar... Eliminar

Variables del sistema

Variable	Valor
ComSpec	C:\Windows\system32\cmd.exe
DriverData	C:\Windows\System32\Drivers\DriverData
HADOOP_HOME	C:\hadoop
NUMBER_OF_PROCESSORS	4
OS	Windows_NT
Path	C:\java\bin;C:\Program Files\Eclipse Adoptium\jdk-21.0.8.9-hotspo...
PATHEXT	.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC

Nueva... Editar... Eliminar

Aceptar Cancelar

Pegar

Opciones de configuración relacionadas

Cambiar la clave de producto para actualizar la edición de Wind

Lee los Términos de licencia y el software de Microsoft

Editar la variable de usuario

Nombre de la variable: Path

Valor de la variable: \ppData\Local\Programs\Microsoft VS Code\bin;C:\Program Files\MongoDB\Server\8.2\bin

Examinar directorio... Examinar archivo...

Aceptar Cancelar

Configuración avanzada del :

Cambiar el nombre de este e (avanzado)

Ayuda de la web

Averiguar cuántos núcleos tie procesador

Comprobar compatibilidad o idiomas

MongoDB

Verificar instalación

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Versión 10.0.19045.6466]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\rdgua>mongod.exe --version
db version v8.2.4
Build Info: {
  "version": "8.2.4",
  "gitVersion": "60692d74054a0bc73f181179e747815825de38ae",
  "modules": [],
  "allocator": "tcmalloc-gperf",
  "environment": {
    "distmod": "windows"
  }
}
```

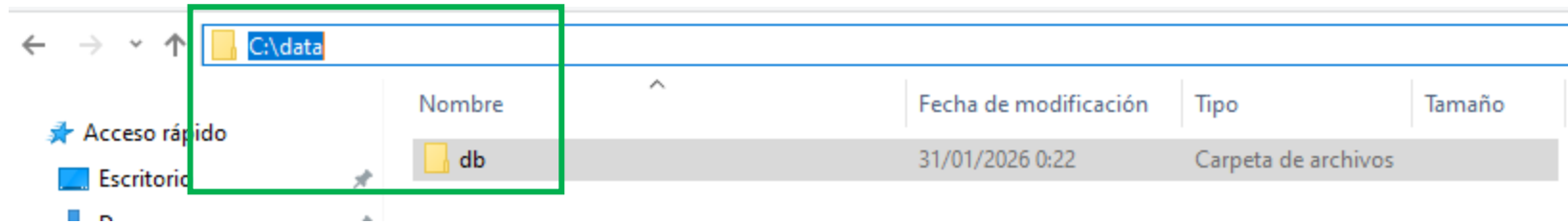
MongoDB

Corregir errores

```
C:\Users\rdgua>
C:\Users\rdgua> mongod
{"t":{"$date":"2026-01-31T00:19:47.984-05:00"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1","msg":"Shuffling initializers","attr":{"seed":3957628804}}
{"t":{"$date":"2026-01-31T00:19:48.003-05:00"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1","msg":"Automatically disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-01-31T00:19:48.094-05:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1","msg":"Initialized wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient":true}}}}
{"t":{"$date":"2026-01-31T00:19:48.095-05:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"Multi threading initialized"}
{"t":{"$date":"2026-01-31T00:19:48.095-05:00"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten","msg":"MongoDB starting","attr":{"pid":13516,"port":27017,"dbPath":"/data/db","architecture":"64-bit","host":"DESKTOP-R6EHI4Q"}}
{"t":{"$date":"2026-01-31T00:19:48.095-05:00"},"s":"I", "c":"CONTROL", "id":23398, "ctx":"initandlisten","msg":"Target operating system minimum version","attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"$date":"2026-01-31T00:19:48.096-05:00"},"s":"I", "c":"CONTROL", "id":23403, "ctx":"initandlisten","msg":"Build Info","attr":{"buildInfo":{"version":"8.2.4","gitVersion":"60692d74054a0bc73f181179e747815825de38ae","modules":[],"allocator":"tcmalloc-gperf","environment":{"distmod":"windows"}}}}}
{"t":{"$date":"2026-01-31T00:19:48.096-05:00"},"s":"I", "c":"CONTROL", "id":51765, "ctx":"initandlisten","msg":"Operating System","attr":{"os":{"name":"Microsoft Windows Workstation (build 19045)","version":"10.0 (build 19045)"}}}
{"t":{"$date":"2026-01-31T00:19:48.097-05:00"},"s":"I", "c":"CONTROL", "id":21951, "ctx":"initandlisten","msg":"Options set by command line","attr":{"options":{}}}
{"t":{"$date":"2026-01-31T00:19:48.098-05:00"},"s":"W", "c":"NETWORK", "id":11621101,"ctx":"initandlisten","msg":"Overriding max connections to honor `capMemoryConsumptionForPreAuthBuffers` settings","attr":{"limit":207995}}
{"t":{"$date":"2026-01-31T00:19:48.099-05:00"},"s":"I", "c":"NETWORK", "id":4648601, "ctx":"initandlisten","msg":"Implicit TCP FastOpen unavailable. If TCP FastOpen is required, set at least one of the related parameters","attr":{"relatedParameters":["tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"]}}
{"t":{"$date":"2026-01-31T00:19:48.102-05:00"},"s":"E", "c":"CONTROL", "id":20557, "ctx":"initandlisten","msg":"DBException in initAndListen, terminating","attr":{"error":"NonExistentPath": "Data directory \\data\\db not found. Create the missing directory or specify another path using (1) the --dbpath command line option, or (2) by adding the 'storage.dbPath' option in the configuration file."}}
{"t":{"$date":"2026-01-31T00:19:48.102-05:00"},"s":"I", "c":"REPL", "id":4784900, "ctx":"initandlisten","msg":"Stepping down the ReplicationCoordinator for shutdown","attr":{"waitTimeMillis":15000}}
{"t":{"$date":"2026-01-31T00:19:48.103-05:00"},"s":"I", "c":"REPL", "id":4794602, "ctx":"initandlisten","msg":"Attempting to enter quiesce mode"}
{"t":{"$date":"2026-01-31T00:19:48.103-05:00"},"s":"I", "c":"-", "id":6371601, "ctx":"initandlisten","msg":"Shutting down the FLE Crud thread pool"}
```

MongoDB

Crear carpeta



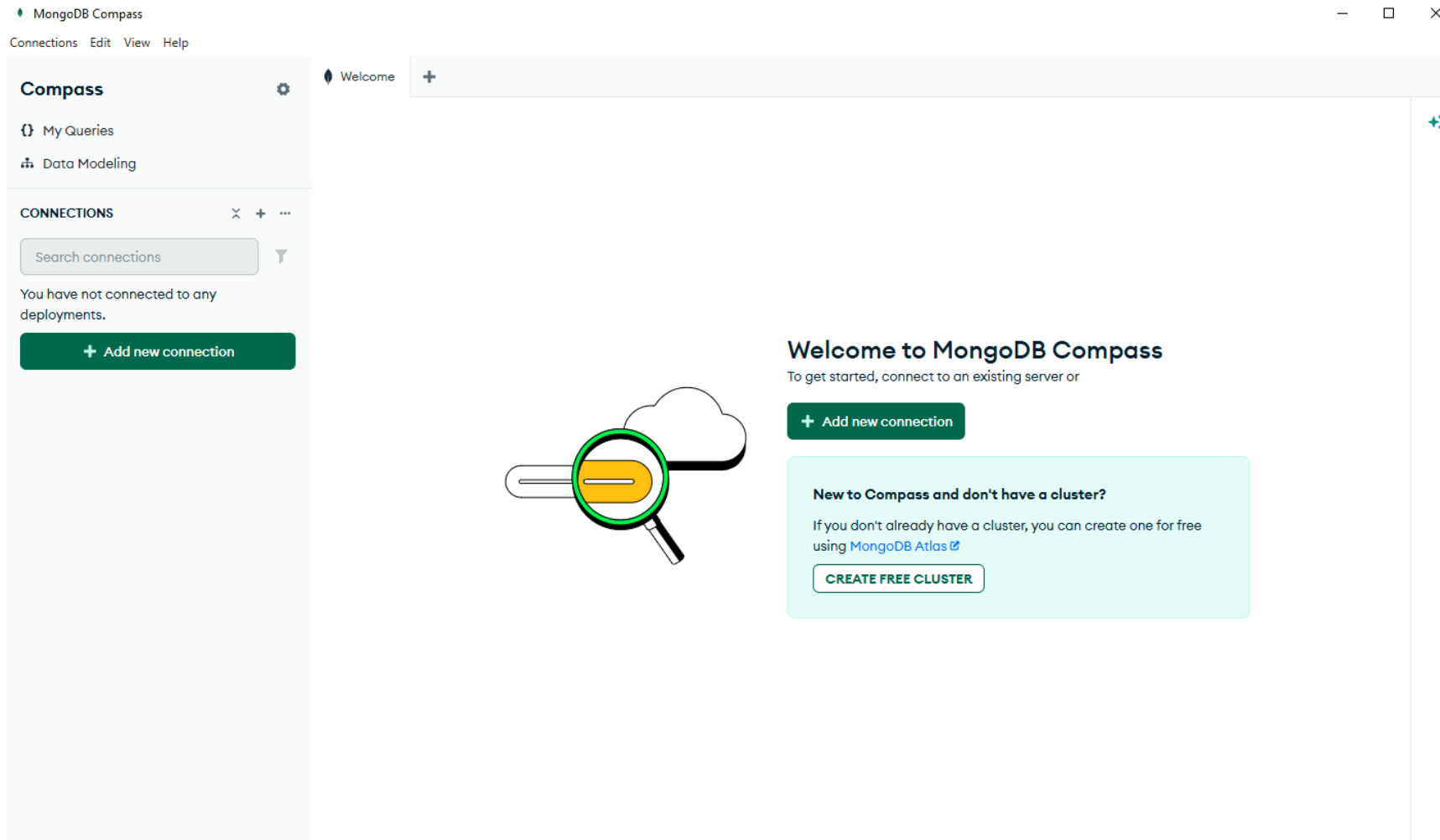
MongoDB

Verificar instalación – Se queda “esperando”

```
ied, oldstate : configrestart {}  
{"t":{"$date":"2026-01-31T00:22:58.844-05:00"},"s":"I", "c":"STORAGE", "id":22262, "ctx":"initandlisten","msg":"Timestamp monitor starting"}  
{"t":{"$date":"2026-01-31T00:22:58.845-05:00"},"s":"I", "c":"STORAGE", "id":7333401, "ctx":"initandlisten","msg":"Starting the DiskSpaceMonitor"}  
{"t":{"$date":"2026-01-31T00:22:58.850-05:00"},"s":"I", "c":"NETWORK", "id":23015, "ctx":"listener","msg":"Listening on","attr":{"address":"127.0.0.1:27017"}}  
{"t":{"$date":"2026-01-31T00:22:58.850-05:00"},"s":"I", "c":"NETWORK", "id":23016, "ctx":"listener","msg":"Waiting for connections","attr":{"port":27017,"ssl"}  
{"t":{"$date":"2026-01-31T00:22:58.851-05:00"},"s":"I", "c":"CONTROL", "id":8423403, "ctx":"initandlisten","msg":"mongod startup complete","attr":{"Summary of t  
lean shutdown?":true,"Statistics":{"setUpPeriodicRunnerMillis":0,"setUpOCSPMillis":0,"setUpTransportLayerMillis":0,"initSyncCrashRecoveryMillis":0,"createLockFile  
tadataMillis":0,"createStorageEngineMillis":179,"writePIDMillis":0,"writeNewMetadataMillis":10,"initializeFCVForIndexMillis":2,"dropAbandonedIdentsMillis":0,"stan  
"userAndRolesGraphMillis":0,"waitForMajorityServiceMillis":1,"startUpReplCoordMillis":1,"recoverChangeStreamMillis":0,"logStartupOptionsMillis":0,"startUpTranspor  
nTotalMillis":1707}}}  
{"t":{"$date":"2026-01-31T00:22:58.853-05:00"},"s":"I", "c":"CONTROL", "id":20712, "ctx":"LogicalSessionCacheReap","msg":"Sessions collection is not set up; w  
ap interval","attr":{"error":"NamespaceNotFound: config.system.sessions does not exist"}}  
{"t":{"$date":"2026-01-31T00:22:58.854-05:00"},"s":"I", "c":"STORAGE", "id":20320, "ctx":"LogicalSessionCacheRefresh","msg":"createCollection","attr":{"namesp  
,"uuidDisposition":"generated","uuid":{"$uuid":"9d194f7f-7f37-4018-93c5-b3b3742aff15"}},"options":{}}}  
{"t":{"$date":"2026-01-31T00:22:58.874-05:00"},"s":"I", "c":"INDEX", "id":20345, "ctx":"LogicalSessionCacheRefresh","msg":"Index build: done building","attr  
onUUID":{"$uuid":"9d194f7f-7f37-4018-93c5-b3b3742aff15"},"namespace":"config.system.sessions","index":"_id_","ident":"index-d555a53a-63c6-4bd9-bd5d-636eb  
collection-424dc849-a534-4204-89f3-28acd3191e37","commitTimestamp":null}}  
{"t":{"$date":"2026-01-31T00:22:58.874-05:00"},"s":"I", "c":"INDEX", "id":20345, "ctx":"LogicalSessionCacheRefresh","msg":"Index build: done building","attr  
onUUID":{"$uuid":"9d194f7f-7f37-4018-93c5-b3b3742aff15"},"namespace":"config.system.sessions","index":"lsidTTLIndex","ident":"index-60925421-ec52-428d-95  
dent":"collection-424dc849-a534-4204-89f3-28acd3191e37","commitTimestamp":null}}
```

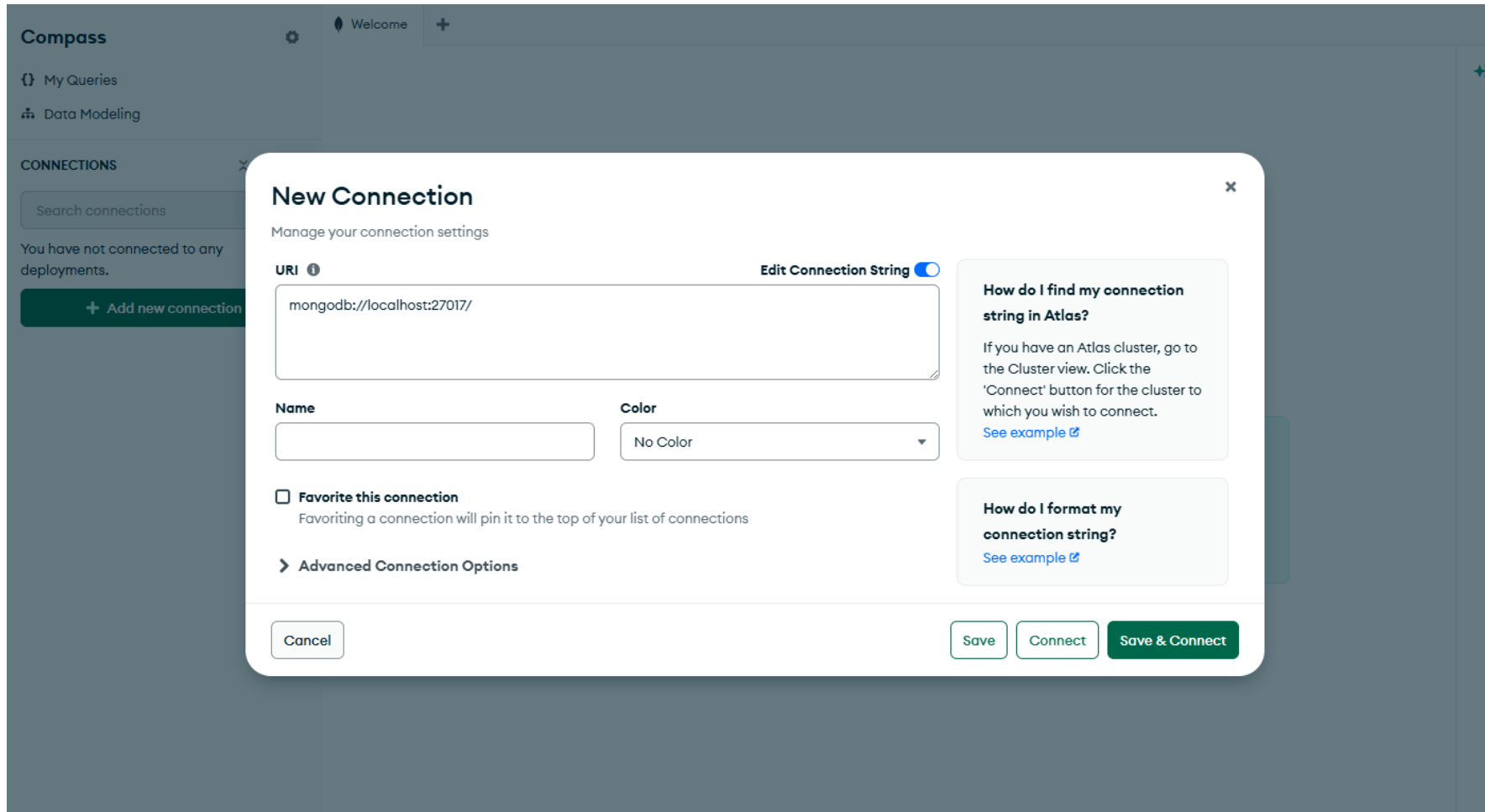
MongoDB

Interacción con Herramienta Compass



MongoDB

Interacción con Herramienta Compass



Gracias Preguntas

Richard Guanoluisa
Bases de Datos II
Pontificia Universidad Católica
del Ecuador